Life Sciences - Grade 10

November 2022 | Section A | Study memo | 50 marks

Generated worked answers for the supplied six-page Answer Booklet. This is an independently prepared study memo, not an official DLSHCCH memorandum.

Suggested marking: follow the printed allocations. Accept equivalent scientific wording. Explanations below support revision; they do not add marks to multiple-choice or matching answers.

Question 1 - Multiple choice [10]

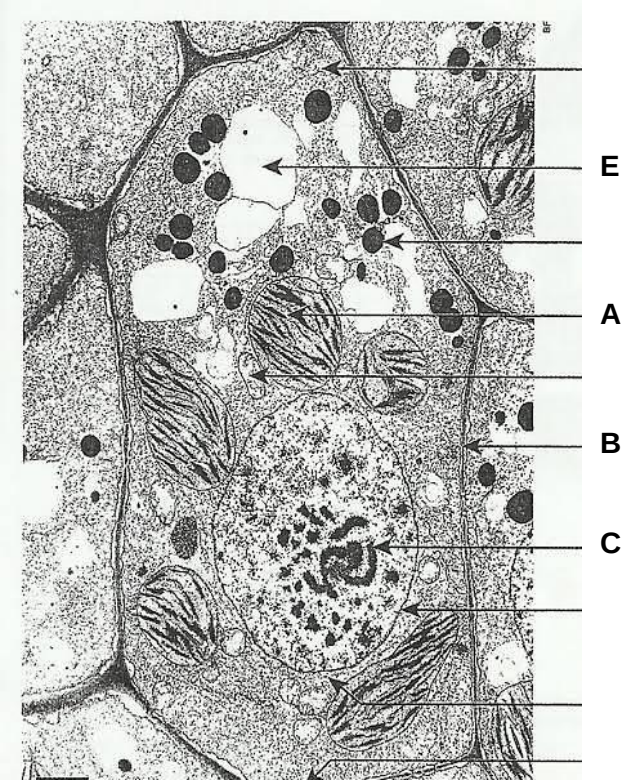
No.	Answer	Explanation	Marks
1.1	C	Respiration transfers energy to ATP: ADP combines with inorganic phosphate to form ATP.	2
1.2	B	Species is the most specific of the four categories listed.	2
1.3	C	The nucleus contains genetic information and regulates cell activities; "cell functioning" is the best option.	2
1.4	B	Root hairs increase the surface area for uptake of water and mineral ions.	2
1.5	C	Facilitated diffusion is passive movement down a concentration gradient through specific membrane proteins.	2

Question 2 - Biological terms [10]

No.	Answer	Explanation	Marks
2.1	Anaphase	Sister chromatids separate and move towards opposite poles.	2
2.2	Cellular respiration	Chemical energy is transferred from food molecules to ATP; energy is not created from nothing.	2
2.3	Granum	One stack of thylakoids. The plural is grana.	2
2.4	Peristalsis	Coordinated waves of muscular contraction move material along the digestive tract.	2
2.5	Food chain	A single feeding pathway showing the direction of energy transfer.	2

Question 3 - Cell micrograph [10]

The letters below correspond to those on page 4 of the supplied booklet. Unlettered arrows are not assessed.



3.1 Plant cell. [2]
A cell wall and chloroplasts support this identification.

3.2 E: vacuole. [2]
The pale compartment indicated by E is a vacuole.

3.3 C: nucleolus function. [2]
Ribosomal RNA (rRNA) is synthesised (1), and ribosomal subunits are assembled (1). Accept "makes ribosomes" as an equivalent concise answer.

3.4 False. [2]
B is the cell/plasma membrane: it is selectively permeable, so not all substances can pass freely. The question asks only for true or false.

3.5 A: chloroplast. [2]
Its internal thylakoid membranes are visible.

Important distinction: C points to the dark nucleolus inside the nucleus. "Controls cell activities" describes the nucleus as a whole, not the nucleolus.

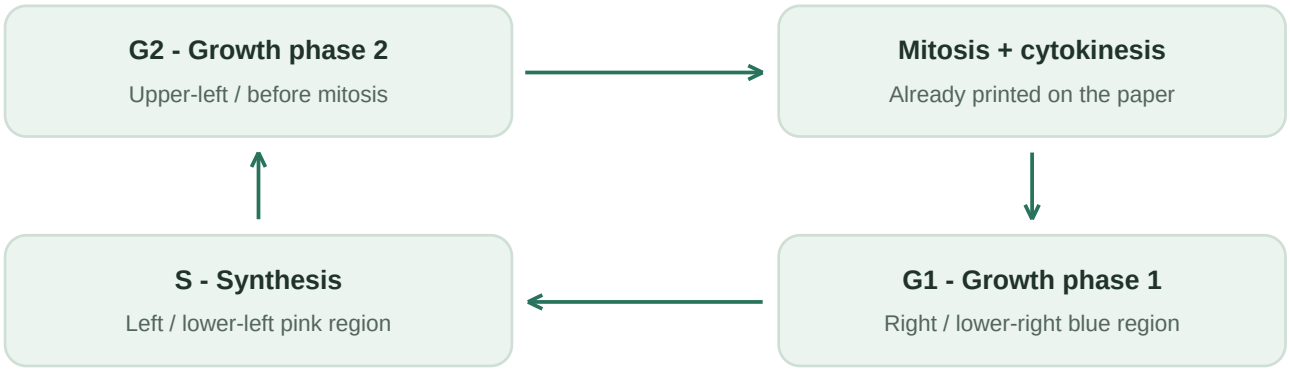
Question 4 - Matching columns [10]

No.	Letter	Meaning	Marks
4.1	A	Habitat: the place or environment in which an organism lives.	2
4.2	E	Biotic: the living components of an ecosystem.	2
4.3	D	Niche: an organism's ecological role and resource use.	2
4.4	G	Stroma: the fluid-filled region of a chloroplast outside the thylakoids.	2
4.5	C	Diffusion: net movement of particles down a concentration gradient.	2

Distractors: B describes the mitochondrial matrix. F refers to stromatolites, not stroma.

Question 5 - Cell cycle [10]

Diagram labels [3]
Award 1 mark for each label placed in the correct region. This location guide matches the regions in the original diagram; box size does not show time spent in a phase.



G1: growth and normal activity after division. S: DNA replication. G2: further growth, checking and preparation before mitosis. G1, S and G2 together form interphase; G0 is not one of the three requested labels.

5.1 Importance of cell regeneration [3]

Award 1 mark for each distinct point, to a maximum of 3:
1. It replaces dead, worn-out or lost cells.
2. It repairs damaged tissues and helps wounds heal.
3. It restores or maintains the normal functioning of tissues and organs.

Accept a clearly explained equivalent biological benefit. Do not award a second mark for simply repeating the same point.

5.2 Phase occupying most of the cell's life [1]

Interphase. (1) This includes G1, S and G2; it is not a stage of mitosis.

5.3 Why the daughter cells are duplicates [3]

1. DNA is replicated during the S phase of interphase. (1)
2. During mitosis, sister chromatids separate and are distributed equally to the two nuclei. (1)
3. Each daughter cell therefore normally receives the same chromosome number and genetic information as the original cell. (1)

Total: Q1 10 + Q2 10 + Q3 10 + Q4 10 + Q5 10 = 50 marks.

Source: supplied DLSHCCH Grade 10 Life Science November 2022 Section A Answer Booklet (6 PDF pages). Organelle terminology cross-check: OpenStax Biology 2e, 4.3 Eukaryotic Cells. Prepared for For Rosa on 3 November 2026.